Day 3 - Documentation

For Day 3,

Get a small recorded NMEA data file.
Understand what GGA and RMC sentences look like.
Try reading those sentences using your ESP32 code.
Extract latitude, longitude, time and GPS status.
Print the result in the Serial Monitor.
it is recorded/simulated data and not live GPS data.

NMEA data : National marine electronics association
A standardized text based communication to allow GPS receivers, marine instruments, and other electronic hardware to share position, speed, and time data.

The GPS module doesn't normally send a ready made JSON object .Instead it sends text sentences

```
$GPRMC,123519,A,1258.000,N,07735.000,E,10.5,180.0,091026,,,A*hh
```

That long string is an NMEA sentence. Each comma-separated field has a particular meaning.

$ - INdicates the beginning of the sentence
GP - Identifies the GPS talker.
GGA - Identifies the kind of information : position,time and fix quality
The comma separated values carry the actual information

What do $GPGGA, $GPRMC and $GNGGA mean?

Sentence	Meaning
$GPGGA	GPS position, time and fix quality
$GPRMC	GPS minimum navigation information
$GNGGA	GGA information from a GNSS talker

$GNRMC	RMC information from a GNSS talker

GGA : Global Positioning System Fix Data
decode every important field
It provides the receiver's time, coordinates, fix quality and related information.

$GPGGA,123519,4807.038,N,01131.000,E,1,08,0.9,545.4,M,46.9,M,,*47

Index	Value	Meaning
0	**GPGGA**	**Message type**
1	**123519**	**Time: 12:35:19 UTC**
2	**4807.038**	**Latitude in NMEA format**
3	**N**	**Northern hemisphere**
4	**1131**	**Longitude in NMEA format**
5	**E**	**Eastern hemisphere**
6	**1**	**Standard GPS fix**
7	**8**	**8 satellites used**
8	**0.9**	**Horizontal dilution of precision (HDOP)**
9	**545.4**	**Altitude value**
10	**M**	**Altitude unit: metres**
11	**46.9**	**Geoid separation**
12	**M**	**Geoid separation unit: metres**

The three fields you should pay special attention to

- Latitude and longitude: tell you where the receiver is.
- Fix quality: tells you whether a usable position fix is available.
- Time: tells you when the position was determined, in UTC.

$GPGGA,123519,4807.038,N,01131.000,E,1,08,0.9,545.4,M,46.9,M,,*47

1. Index 0 — GPGGA

This identifies the message type.

- GP = GPS talker ID.
- GGA = position fix data.

It tells your ESP32 parser what kind of information the sentence contains.

2. Index 1 — 123519

This is the time in UTC.

Split it as:

- 12 = hours
- 35 = minutes
- 19 = seconds

So, the time is 12:35:19 UTC. It is not necessarily Indian Standard Time (IST).

3–4. Index 2 and 3 — 4807.038,N

This represents latitude.

- 48 = degrees
- 07.038 = minutes
- N = North

Convert it to decimal degrees:

$$48 + \frac{7.038}{60} = 48.1173^{\circ}$$

5–6. Index 4 and 5 — 01131.000,E

This represents longitude.

- 011 = degrees
- 31.000 = minutes
- E = East

$$11 + \frac{31}{60} = 11.516667^{\circ}$$

Why convert? Because map APIs and databases commonly use decimal degrees.

7. Index 6 — 1

This is the fix quality.

Value	Meaning
0	Invalid fix
1	Standard GPS fix
2	Differential GPS fix

Other values may represent additional positioning modes. The exact supported modes depend on the receiver.

8. Index 7 — 08

This means 8 satellites were used to calculate the position.

More satellites can help, but the number alone does not guarantee accuracy.

9. Index 8 — 0.9

This is HDOP: Horizontal Dilution of Precision.

It describes how satellite geometry affects horizontal positioning precision.

- Lower HDOP generally means better satellite geometry.
- Higher HDOP generally means worse satellite geometry.

It is not an error value in metres.

10–11. Index 9 and 10 — 545.4,M

This is altitude:

- 545.4 = altitude value
- M = metres

For this example, the reported altitude is 545.4 metres above mean sea level.

12–13. Index 11 and 12 — 46.9,M

This is geoid separation.

- 46.9 = separation value
- M = metres

The geoid is a reference surface related to mean sea level. Geoid separation helps relate the GNSS ellipsoid-based height to the mean-sea-level-based altitude.

You don't need this field for your initial vehicle-tracking task.

14–15. Index 13 and 14 — empty fields

The sentence ends this section with:

,,*47

The empty fields mean no values were supplied for those fields. Empty does not mean zero.

Final field — *47

This is the checksum.

The receiver or parser uses it to detect certain transmission errors. A checksum does not guarantee that the coordinates themselves are accurate.

- A 1 in the GGA fix-quality field indicates a standard GPS fix. A 0 indicates that the fix is invalid. Other values can represent differential or specialised positioning modes, depending on the receiver.
- One subtle point: eight satellites used does not automatically mean the location is highly accurate. HDOP provides information about satellite geometry; it is not itself an error measurement in metres.

RMC: decode every important field

RMC stands for Recommended Minimum Specific GNSS Data.

$GPRMC,081836,A,3751.65,S,14507.36,E,000.0,360.0,130998,011.3,E*62

It is especially useful for your vehicle-tracking project because it includes position, time, validity, speed and course.

Two fields matter a lot for your firmware:

- A means the navigation data is valid; V means void or invalid.
- Speed is normally reported in knots, not kilometres per hour. One knot is 1.852 km/h. Course is measured in degrees relative to true north.

NMEA coordinates : GPS coordinates in NMEA sentences use degrees and decimal minutes, not the decimal-degree format commonly used in maps and databases

48 = degrees **07.038 = minutes** **N = north**

$$Decimaldegrees = Degrees + \frac{Minutes}{60}$$

$$48 + \frac{7.038}{60} = 48.1173^{\circ}$$

Index	Value	Meaning
0	**GPRMC**	**Message type**
1	**81836**	**Time: 08:18:36 UTC**
2	**A**	**Data status: valid**
3	**3751.65**	**Latitude in NMEA format**
4	**S**	**Southern hemisphere**

5	**14507.36**	**Longitude in NMEA format**
6	**E**	**Eastern hemisphere**
7	**0**	**Speed over ground in knots**
8	**360**	**Course over ground in degrees**
9	**130998**	**Date: 13 August 1998**
10	**11.3**	**Magnetic variation**
11	**E**	**Magnetic variation direction**
—	***62**	**Checksum**

What is a checksum :

...,,*47

*** marks the beginning of the checksum field.** **47 is a hexadecimal checksum value.**

- A checksum is used to detect certain transmission errors. In the common NMEA 0183 format, the checksum is calculated by XOR-ing the characters between $ and *, excluding those two delimiters.

- Imagine the receiver sends a character incorrectly because of a communication problem. A checksum check can reveal that the sentence may have been corrupted.
- However, a valid checksum does not prove that the GPS location is accurate. It only checks the integrity of the sentence's contents.

For a physical connection, the usual wiring is:

- GPS TX → ESP32 UART RX
- GPS RX → ESP32 UART TX, if sending commands to the GPS is required
- GPS GND ↔ ESP32 GND

